

Planar Kepler Conics, Collision Incompleteness, and the Levi–Civita Configuration Boundary

Route-A structural certificate HCS-C216

Abstract

We close a convention-locked theorem package for the attractive planar Kepler Hamiltonian. Runge–Lenz identities classify every nonzero-angular-momentum orbit as an ellipse, parabola, or hyperbola; the negative-energy period and radial action and the positive-energy scattering angle are normalized exactly. The zero-angular-momentum branch reaches the removed collision in finite physical time. Levi–Civita’s fixed-energy change of variables gives a smooth configuration equation, while a time-period resonance fixes a continuum rather than isolated primitive orbits. The radial action is written both as a closed-cycle integral and as its turning-point half-cycle, and all three collision antiderivatives are displayed. An executable exact ledger and independent symbolic checks delimit the result: the configuration continuation is not a full Ligon–Schaaf symplectomorphism, and the strict Route-A arithmetic gates stop.

Keywords: Kepler problem; Runge–Lenz vector; collision regularization; radial action; fixed-point continuum; Route A.

1 Frozen Hamiltonian and invariants

Identify the plane with $\mathbb{C}$ only when the Levi–Civita map is introduced. For $q \in \mathbb{R}^2 \setminus \{0\}$, $p \in \mathbb{R}^2$, and $\mu > 0$, set

$$H(q, p) = \frac{|p|^2}{2} - \frac{\mu}{r}, \quad r = |q|.$$

Hamilton’s equations are $\dot{q} = p$ and $\dot{p} = -\mu q/r^3$. Along a collision-free solution define

$$E = H, \quad L = q_1 p_2 - q_2 p_1, \quad A = (|p|^2 - \mu/r)q - (q \cdot p)p. \quad (1)$$

Differentiation gives $\dot{E} = \dot{L} = \dot{A} = 0$. Taking the scalar product and norm of A gives the two identities

$$A \cdot q = L^2 - \mu r, \quad |A|^2 = \mu^2 + 2EL^2. \quad (2)$$

2 All-energy conics, period, action, and scattering

Suppose $L \neq 0$, let $e = |A|/\mu$, and measure θ from A . Equation (2) is equivalent to

$$r(\theta) = \frac{L^2/\mu}{1 + e \cos \theta}, \quad e^2 - 1 = \frac{2EL^2}{\mu^2}. \quad (3)$$

Thus $E < 0$ is an ellipse (a circle when $e = 0$), $E = 0$ a parabola, and $E > 0$ a hyperbola. In the bound case put $a = -\mu/(2E)$ and $r_{\pm} = a(1 \pm e)$. Kepler’s period and the radial momentum are

$$P(E) = 2\pi\mu(-2E)^{-3/2}, \quad p_r^2 = 2\left(E + \frac{\mu}{r}\right) - \frac{L^2}{r^2}. \quad (4)$$

The normalization is part of the theorem, not a convention left implicit:

$$J_r = \frac{1}{2\pi} \oint p_r \, dr = \frac{1}{\pi} \int_{r_-}^{r_+} p_r \, dr = \frac{\mu}{\sqrt{-2E}} - |L|, \quad E < 0. \quad (5)$$

The turning-point substitution also gives $\partial_E J_r = P(E)/(2\pi)$. For $E > 0$, the asymptotic directions of (3) have $\cos \theta_{\infty} = -1/e$, so the unsigned hyperbolic deflection is

$$\chi = 2 \arcsin(1/e), \quad e > 1. \quad (6)$$

3 The radial collision boundary

When $L = 0$, the inward branch obeys

$$\dot{r}^2 = 2(E + \mu/r), \quad t_{\text{coll}}(r_0) = \int_0^{r_0} \frac{dr}{\sqrt{2(E + \mu/r)}} < \infty, \quad (7)$$

for every admissible starting radius. The endpoint $r = 0$ is absent from the physical phase space, hence the physical flow is incomplete. With $\alpha = -E > 0$, $u = \arcsin \sqrt{\alpha r_0/\mu}$ gives

$$t_{\text{coll}} = \frac{\mu}{\sqrt{2\alpha^{3/2}}} (u - \sin u \cos u);$$

for $E = 0$ it is $2r_0^{3/2}/(3\sqrt{2\mu})$, while for $E > 0$, with $u = \text{arsinh} \sqrt{Er_0/\mu}$, it is

$$t_{\text{coll}} = \frac{\mu}{\sqrt{2E^{3/2}}} (\sinh u \cosh u - u). \quad (8)$$

These formulas separate the collision boundary from the $L \neq 0$ conic classification; no collision is silently counted as a periodic orbit.

4 Levi–Civita equation and what is (not) extended

Write $q = u^2$ with $u \in \mathbb{C}$, and set $dt = |u|^2 d\tau$. On a fixed energy level, direct substitution gives

$$u'' = \frac{E}{2}u, \quad 2|u'|^2 - E|u|^2 = \mu, \quad L = 2\operatorname{Im}(\bar{u}u'). \quad (9)$$

The equation in τ is smooth through $u = 0$, so (9) supplies a configuration-level collision continuation. The physical time density then vanishes at the collision. We make no claim of a global canonical transformation, a full Ligon–Schaaf symplectomorphism, or a three-dimensional regularization theorem.

5 Fixed-time resonance and strict stop

Every collision-free ellipse at energy $E < 0$ has the same period $P(E)$. Consequently a time- T map has the fixed continuum

$$\{H = E, L \neq 0\} \quad \text{whenever} \quad T = mP(E), \quad m \in \mathbb{N}. \quad (10)$$

This set has dimension three in the four-dimensional phase space, so ordinary isolated Artin–Mazur primitive-orbit counting is undefined on the resonant shells. Parabolic and hyperbolic noncollision trajectories escape and have no finite period.

The release ledger has 10 exact orbit rows, 4 radial rows, 12 Levi–Civita rows, and 5 fixed-set rows. The producer-independent checker passes 260 assertions; the SymPy cross-check passes 17 identities; replay is byte exact; and 25 hostile mutations are rejected (24 repaired-hash plus one stale-hash, including an unknown-key injection). These finite rows are regression evidence only.

The source-attributed classical references are Levi–Civita [LC1920], Moser [M70], and Ligon–Schaaf [LS76]; no priority claim is made. With scope `NO_BAD_EULER_OR_ROOT_NUMBER`,

$$(A0, A1, A2, A3, A4) = (\text{FAIL}, \text{WEAK}, \text{FAIL}, \text{FAIL}, \text{NATURAL QUANTIZATION}),$$

overall `ROUTE_A_REJECTED`; Route B is false. The natural Coulomb/oscillator quantization hint is not an operator construction. No target primes or zeros, arithmetic local data, Euler factors, root numbers, automorphy, target functional equation, or Hilbert–Pólya operator is used or claimed.

References

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- [2] J. Moser, “Regularization of Kepler’s problem and the averaging method on a manifold,” *Communications on Pure and Applied Mathematics* 23 (1970), 609–636. DOI: 10.1002/cpa.3160230406.
- [3] T. Ligon and M. Schaaf, “On the global symmetry of the classical Kepler problem,” *Reports on Mathematical Physics* 9 (1976), 281–300. DOI: 10.1016/0034-4877(76)90061-6.

Revision focus. Round 2 adds Levi–Civita scope, integer strobe resonance, executable counts, source attribution, and the strict Route-A stop.

Declarations

Data and code. Synthetic exact probes and package-local deterministic code accompany the manuscript; no training data are used. **Ethics/funding.** No human or animal data; no external funding reported. **AI disclosure.** An AI coding assistant supported drafting and exact-code development; it was not an external reviewer or independent error process.